CURRICULUM VITAE

Jiye Kim

lool1015@snu.ac.kr

Updated in June 2026

EDUCATIONS

2020.09 - Present	Ph. D. candidate in Electrical and Computer Engineering Department of Electrical and Computer Engineering **Seoul National University**, Seoul, Korea *Adviser: Professor Jongho Lee*
2025.11 - 2026.04	Visiting Student **Athinoula A. Martinos Center for Biomedical Imaging**, Massachusetts General Hospital and Harvard Medical School Charlestown, USA *Adviser: Professor Berkin Bilgic*
2016 – 2020.08	B.S. in Electrical and Computer Engineering Department of Electrical and Computer Engineering **Seoul National University**, Seoul, Korea *Cum Laude Honors*
2019	Study Abroad Exchange Program, Electrical Engineering **ETH Zurich**, Zurich, Switzerland

HONORS AND AWARDS

- ISMRM Magna Cum Laude Merit Award, “Simultaneous reconstruction of longitudinal QSM data (Longitudinal-QSM)”, 2026

- 2nd Place Abstract Award, ISMRM EMTP Study Group, “Simultaneous reconstruction of longitudinal QSM data (Longitudinal-QSM)”, 2026

- Best Oral Presentation 1st, ISMRM EMTP Study Group Trainee Day, “Simultaneous reconstruction of longitudinal QSM data (Longitudinal-QSM)”, 2026

- ISMRM Magna Cum Laude Merit Award, “Visualization of in-vivo brainstem myeloarchitecture via χ-separation at 7T”, 2025

- Best Trainee Scientific Awards Poster Presentation Grand Prix, International Congress on MRI & Annual Scientific Meeting of KSMRM, 2023

- Best Trainee Scientific Awards Poster Presentation Gold, International Congress on MRI & Annual Scientific Meeting of KSMRM, 2022

- Best Trainee Scientific Awards, Oral Presentation 2nd, International Congress on MRI & Annual Scientific Meeting of KSMRM, 2021

- The Encouragement Prize, The Joint Conference of the IBEC and the ICBHI, 2021

- Academic Excellence Scholarship, 2018-2019

- Global Exchange Scholarship, Mirae Asset Park Hyeon Joo Foundation, 2019

Journal Publications

- T. Kim, S. Ji, K. Min, M. Kim, J. Youn, C. Oh, **J. Kim**, and J. Lee, "Vessel segmentation for χ-separation", *Magnetic Resonance in Medicine* (2025)

- Y. Chen, Y. Jun, A. Heydari, X. Yong, **J. Kim**, J. Lee, ... & Bilgic, B., "MIMOSA: Multi-parametric Imaging using Multiple-echoes with Optimized Simultaneous Acquisition for highly-efficient quantitative MRI", *Magnetic Resonance in Medicine* (2025)

- C. Oh, W. Jung, **J. Kim**, H. Jeong, and J. Lee, "A fair comparison in deep learning-based QSM reconstruction", *IEEE Access* (2025)

- **J. Kim**, M. Kim, S. Ji, K. Min, H. Jeong, H.-G. Shin, C. Oh, R. J. Fox, K. E. Sakaie, M. J. Lowe, S.-H. Oh, S. Straub, S.-G. Kim, and J. Lee, "In-vivo high-resolution χ-separation at 7T", *NeuroImage* (2025)

- M. Kim, S. Ji, **J. Kim**, K. Min, H. Jeong, J. Youn, T. Kim, J. Jang, B. Bilgic, H.-G. Shin, and J. Lee, "χ-sepnet: Deep neural network for magnetic susceptibility source separation", *Human Brain Mapping* (2025)

- D. Shin, Y. Kim, C. Oh, H. An, J. Park, **J. Kim**, and J. Lee, "Deep Reinforcement Learning-Designed Radiofrequency waveform in MRI," *Nature Machine Intelligence* (2021)

Under Review

- **J. Kim**, H. Jeong, T. Kim, E. Jeong, J. Jang, Y. Choi, and J. Lee, "Longitudinal QSM: Enhancing Consistency of Multiple Time Point Susceptibility Maps via Simultaneous Reconstruction"

- E. Zhang, Y. Mori, **J. Kim**, H. Lee, J. Lee, and T. Hanakawa, "Paramagnetic and Diamagnetic MRI Markers of Cognitive Decline in Aged Population explored at 7T"

- Y. Chen, Y. Jun, H. G. Shin, S. Li, S. Fujita, X. Yong, **J. Kim**, et al., “MWF-MIMOSA for Efficient Simultaneous Relaxometry and Myelin Water Fraction Mapping”

Conferences

Oral

- **J. Kim**, H. Jeong, T. Kim, Y. Choi, and J. Lee, “Simultaneous reconstruction of longitudinal QSM data (Longitudinal-QSM)”, 2026 ISMRM Annual Meeting & Exhibition

- **J. Kim** and J. Lee, “DeepRF-Grad: Simultaneous Design of RF Pulse and Slice Selective Gradient Using Reinforcement Learning”, 2021 International Congress on MRI & Annual Scientific Meeting of KSMRM

Poster

- **J. Kim**, C. Oh, A.-M. Oros-Peusquens, N. J. Shah, C.-H. Shon, H. Song, and J. Lee, “Visualization of in-vivo brainstem myeloarchitecture via χ-separation at 7T.”, 2025 ISMRM Annual Meeting & Exhibition

- **J. Kim**, H.-G. Shin, M. Kim, S. Ji, K. Min, H. Jeong, S.-G. Kim, and J. Lee, “In-vivo high-resolution χ-separation (chi-separation) at 7T”, 2024 ISMRM Annual Meeting & Exhibition

- **J. Kim** and J. Lee, “Simultaneous Multislice Pulse Optimization Using DeepRF”, 2023 International Congress on MRI & Annual Scientific Meeting of KSMRM

- **J. Kim**, H.-G. Shin, S. Ji, H. Jeong, H. An, C.-H. Sohn, S. Han, H. Song, J.-H. Chae, S.-J. Choi, J.-J. Sung, and J. Lee, “A feasibility study of susceptibility source separation via chi-separation in amyotrophic lateral sclerosis patients at 7T”, 2023 ISMRM Annual Meeting & Exhibition

- **J. Kim**, H. An, C. Oh, B. Bilgic, and J. Lee, “DeepRF for simultaneous multislice pulses”, 2023 ISMRM Annual Meeting & Exhibition

- **J. Kim** and J. Lee, “Accelerated DeepRF using optimal control”, 2022 International Congress on MRI & Annual Scientific Meeting of KSMRM

- **J. Kim**, D. Shin, H. An, H. Jeong, M. Kim, and J. Lee, “Accelerated DeepRF using modified optimal control”, 2022 ISMRM Annual Meeting & Exhibition

- **J. Kim**, D. Shin, J. Park, H. Jeong, and J. Lee, “DeepRF-Grad: Simultaneous design of RF pulse and slice selective gradient using self-learning machine.” 2021 ISMRM Annual Meeting & Exhibition

Teaching Experiences

- Teaching Assistant, Bioelectrical Information Engineering (undergraduate course), Fall semester, 2023, Seoul National University, Korea

- Teaching Assistant, Signals and Systems (undergraduate course), Spring semester, 2021, Seoul National University, Korea

- Teaching Assistant, Topics in Communications (graduate course), Spring semester, 2020, Seoul National University, Korea

Professional Experiences

- ISMRM & ISMRT Annual Meeting & Exhibition, 07-12 May 2022, London, UK.
  - *Moderator*, Oral Session: RF Pulse Design, Parallel Transmission & B1 Shimming

Skills

- Programming: MATLAB, Python, Linux

- MRI Systems: SIEMENS 3T and 7T operation

- MRI Sequence Development: Pulseq-based sequence programming

- NeuroImaging Tools: FSL, ANTs, SPM

- Machine Learning: Pytorch, Tensorflow